

Driving Through the Data: Extended Reality Perceptual Support System (XRPSS)

Proof-of-concept for spatial cognition support during and after off-road driving

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ABSTRACT

This work introduces the Extended Reality Perceptual Support System (XRPSS) that provides real-time 3D LiDAR sensor data to off-road vehicle drivers with interactive XR for enhancing in-situ spatial awareness in natural environments. The system extends to ex-situ data exploration via motion driving simulators and VR. We created a proof-of-concept system that can both visualize real-time LiDAR point clouds in the field and generate virtual experiences for the motion simulator post-drive.

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1 CONCEPT

In this work, we present the Extended Reality Perceptual Support System (XRPSS) - a proof-of-concept interactive visualization of real-time 3D LiDAR data during off-road driving, allowing users to query the environment. The XRPSS has two distinct functions. First, it makes 3D LiDAR point cloud available to operators in the vehicle in real-time when encountering challenging driving scenarios. Second, it generates 3D maps of natural environments for ex-situ study, which can be viewed in-vehicle, stored or sent over network to other users.

In off-road vehicle operations, a ‘spotter’ assists the driver from outside of the vehicle to provide an alternative perspective on the environment. Inspired by spotters, the XRPSS is intended for scenarios that call for more advanced spatial cognition. For ex-situ analysis of spatial data, we introduced a driving simulator for exploring data. The XRPSS’ design creates spatial congruence between the physical reality of the vehicle and the data visualization space. By simulating driving through the data, we match the perceptual



Figure 1: Example scenario where the driver may benefit from tools to enhance their spatial awareness.

conditions between data collection and analysis. From our understanding, this is the first proof-of-concept combining real-time 3D LiDAR, emerging interfaces, and exploration of natural terrain.

2 BACKGROUND

Ground vehicle-mounted LiDAR has been used for mapping urban environments [WANG et al., 2019]; for navigating autonomous vehicles [LIU et al., 2018]; and for generating datasets of forests [PIERZCHAŁA et al., 2018]. Extended Reality (XR) enables deeper interactivity with geospatial data [ÇÖLTEKIN et al., 2020], such as exploration of coastlines [HEDLEY and LOCHHEAD, 2020] and the seafloor [LOCHHEAD and HEDLEY, 2021]. Vehicle manufacturers are using XR to deliver unique experiences for driver training [GHIURĂU et al., 2020] and test drives [LATZA, 2022]. Researchers outlined a framework for XR for drivers: tasks, interactions, and interfaces [de Oliveira FARIA, 2020; GABBARD et al., 2014].

3 METHODS

This work provides a configuration framework, which wherever possible are open-source, free, and off-the-shelf, to facilitate replication of the analyses. The XRPSS is open-source and accessible here: <https://github.com/spatial-interface-research-lab/3D-mapping-system.git>.

The Robot Operating System (ROS) standardizes the communications protocol across different developers of robotics hardware and software [QUIGLEY et al., 2009]. ROS served as the nervous system between the LiDAR sensor, mapping algorithms, and the Unity Game Engine. We chose the Ouster OS0-32 LiDAR to balance cost and capabilities. We designed and constructed the mount with 2020 extrusion and braces made from laser cut acrylic sheet. The

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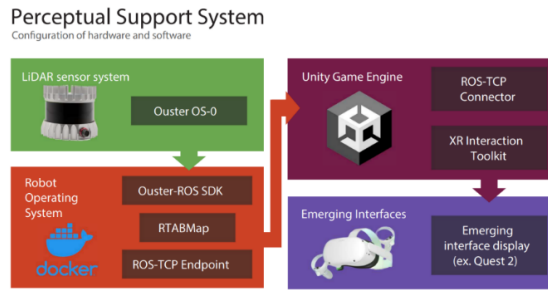


Figure 2: Diagram showing the configuration of technologies in the Perceptual Support System, including the LiDAR sensor system, Robot Operating System, Unity Game Engine, and Emerging Interfaces.

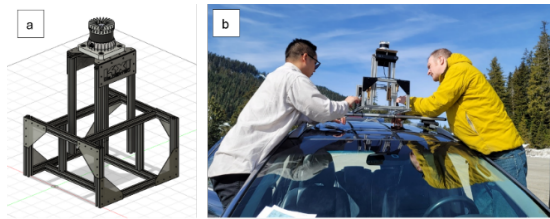


Figure 3: (a) The LiDAR sensor and the adjustable mount is shown as visualized in the AutoCAD Fusion 360 software in which it was designed. (b) The built hardware is attached to a test vehicle by researchers.

Ouster-ROS SDK allows the data from the Ouster OS0-32 unit to be converted to a standardized ROS message type. We used the Adafruit Ultimate GPS Featherwing as an input on the vehicle position to allow maps to have a global coordinate system. Real-Time Appearance-Based Mapping (RTAB-Map) is a Simultaneous Localization And Mapping approach that creates 3D point clouds from LiDAR inputs [LABBÉ and MICHAUD, 2019]. We chose RTAB-Map as it is regularly updated, open-source, and compatible with ROS. For the visualization, we utilized the Unity game engine with the ROS-TCP Endpoint and ROS-TCP Connector Unity packages. The XR Interaction Toolkit package in Unity allows our visualizations to be in XR [Unity, 2023]. We used the Meta Quest 2 for prototyping and testing. We can scale, rotate, translate the point cloud generated.

4 IMPROVEMENTS ON EXISTING METHODS

LiDAR is more precise than other real-time sensors, like stereo cameras, with further range. With live-monitoring of data feeds, the operator can make decisions in the field to increase the quality of the data. The complexity of LiDAR data makes it difficult to understand in real-time through 2D displays, while 3D spatial data comparisons are underexplored [KIM et al., 2017]. We explored 3D for faster spatial cognition by users and linking data to reality. We can answer questions like “can my vehicle fit through this path?” as the vehicle itself becomes a tool for querying space and data. Our system combines mapping with hazard avoidance using the same sensor system. We designed our system to be modular to include additional sensor systems, explore interface permutations, and enable connectivity for novel experiences at distance.

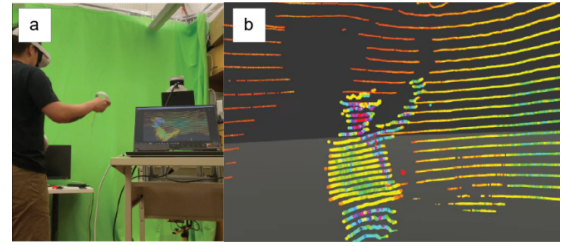


Figure 4: The proof-of-concept Perceptual Support System in the lab environment, where (a) shows the configuration of the LiDAR in the lab and (b) is the user's perspective of the point cloud visualization in VR.

5 NEXT STEPS

We would like to integrate Passthrough VR with a 360 degree camera. We could streamline controls with integration into the steering wheel, as the user currently juggles many different controls. We will conduct empirical studies to collect user feedback. The intention of this work is for field use, so we need to test the XRPSS with off-road drivers in challenging driving conditions.

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